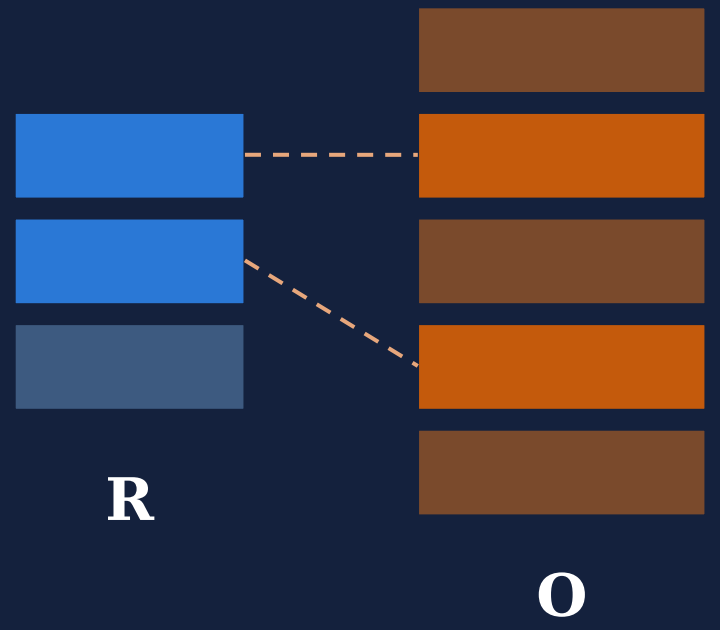
A worked example of the protocol

Three cohort records, five register records: every step of the protocol with its numbers



Companion to the introduction deck · October 2026

How to read this example

Real steps, toy sizes

Every phase of the protocol, in order, with the numbers each step produces.

Shown in the clear

We print values the protocol keeps encrypted, so you can check each step. A party only ever sees what its own boxes say.

Too small to be useful

Two people match and the model has five coefficients: least squares has no unique answer, and the privacy noise dwarfs the data. Watch how the protocol fails safe; the last slides repeat the steps with 20,000 people.

Notation

I: R's cohort members found in O's register. π: the bijection sending each $i \in I$ to its row in O's register. τ: O's secret shuffle of its rows. Budget: $\varepsilon = 1$, $\delta = 10^{-9}$.

The data

Already scaled so that every value lies in $[-1, 1]$

Researcher R: 3 records × 2 features

id	education x_{R1}	health x_{R2}
104	0.5	−0.2
221	−0.4	0.8
317	0.9	0.1

Organization O: 5 records × 2 features, plus y

id	age x_{O1}	benefits x_{O2}	income y
088	0.3	1.0	−0.5
104	−0.6	1.0	0.7
150	0.1	1.0	−0.2
221	0.8	0.0	0.4
399	−0.2	0.0	0.9

Goal: regress income on education, health, age and benefits, plus an intercept, over the people in both tables. Neither party knows yet who that is: identifiers 104 and 221 appear in both, 317 only in R's cohort.

The answer we are after

Computed here by pooling the data, only to check the protocol's steps against it

Matched set I = {104, 221}, so n = 2. Rows of $Z = [X_R^I \mid X_O^I]$ and y^I:

	intercept	education	health	age	benefits	income
104	1.0	0.5	−0.2	−0.6	1.0	0.7
221	1.0	−0.4	0.8	0.8	0.0	0.4

Gram matrix $G^I = Z^T Z$

	intercept	education	health	age	benefits
intercept	2.00	0.10	0.60	0.20	1.00
education	0.10	0.41	−0.42	−0.62	0.50
health	0.60	−0.42	0.68	0.76	−0.20
age	0.20	−0.62	0.76	1.00	−0.60
benefits	1.00	0.50	−0.20	−0.60	1.00

$c^I = Z^T y^I$

c
1.10
0.19
0.18
−0.10
0.70

$y^T y = 0.65$

Read the colours as in the introduction

- Blue A^I: R's rows only
- Light orange B^I, c_O^I: O's matched rows
- Orange C^I, c_R^I: pairs R's row i with O's row π(i)
- **G has rank 2 < 5: with two people, least squares has no unique solution. The ridge will matter.**

Phase 0: setup

Phase 0 · R + O

Everything that sets the noise is public and fixed before any data are touched

Agreed in public

- Caps: R pads its identifiers to $n_R^{max} = 4$; O pads its register to $\bar{n}_O = 6$
- Bounds: every value in $[-1, 1]$, so $B_R = \sqrt{3}$ (two features and the intercept), $B_O = \sqrt{2}$, $B_y = 1$
- Budget $\varepsilon = 1$, $\delta = 10^{-9}$; stage-1 share $f_1 = 0.1$; target $\vartheta^* = 2.5$; $\alpha = 0.05$
- Split key: $s = s_R + s_O$, neither share known to the other party

Derived by O

Sensitivity	$\Delta^2 = (2{\cdot}2 + 3 + 1)^2/2 + 2{\cdot}3{\cdot}1 = 38$	$\Delta = 6.16$
Stage 1 only (B, C)	$\Delta_G = (2{\cdot}2 + 3)/\sqrt{2}$	$\Delta_G = 4.95$
Budget	tight zCDP conversion of $(1, 10^{-9})$	$\rho_{run} = 0.0150$
Stage-1 noise	$\Delta_G / \sqrt{(2 \cdot 0.1 \cdot \rho_{run})}$	$\sigma_1 = 90.5$
Stage-2 noise	$\Delta / \sqrt{(2 \cdot 0.9 \cdot \rho_{run})}$	$\sigma_2 = 37.5$

Already visible: each released number will carry noise of about ±38, while the statistics are about 1. With two people, privacy must win.

Phase 1: link

Phase 1 · R + O

Labeled PSI: R learns which of its identifiers match, and where those rows sit in O's shuffled register

R's padded identifiers

id	F_k(id)	result
104	20938	**match → position 6**
221	39362	**match → position 3**
317	12628	no match
9d1 (dummy)	58249	no match

O's padded register, after its secret shuffle τ

id	F_k(id)	position τ	note
088	40782	4	
104	20938	6	
150	35815	1	
221	39362	3	
399	52946	5	
7e5 (dummy)	35081	2	$x_O = 0$, $y = 0$

R learns: π(104) = 6, π(221) = 3; 317 is not in the register.

O learns: 4 queries and its own 6 rows; not which matched, nor that R has 3 people.

F_k(id): O's keyed hash (OPRF), worked out with toy numbers on the next three slides.

Phase 1 by hand: the OPRF

Phase 1 · R ↔ O

Toy group: squares mod p = 1019, of prime order q = 509; O's key k = 433; no security

1. R blinds: $\alpha = h^r \bmod p$, with $h = H_1(\text{id})$, r random
2. O raises to its key: $\beta = \alpha^k \bmod p$
3. R unblinds: $h^k = \beta^{1/r} \bmod p$, 1/r taken mod q

id	$h = H_1(\text{id})$	r	$\alpha = h^r$	$\beta = \alpha^k$	$1/r \bmod q$	$h^k = \beta^{1/r}$	item $F_k(\text{id})$	label key $K(\text{id})$
104	819	92	348	286	83	516	**20938**	**60918**
221	291	15	109	1011	34	154	**39362**	**39074**
317	956	326	494	144	89	991	**12628**	**29997**
9d1 (dummy)	672	187	845	616	49	64	**58249**	**42829**

O never sees the ids

It sees only α = 348, 109, 494, 845. Because r is random, each could be any element of the group. O makes exactly $n_R^{max} = 4$ evaluations.

Check

O knows k, so for its own rows it computes h^k directly: $H_1(104)^{433} \bmod 1019 = 516$, the value R unblinded. Same h^k, same item: 104 can match.

Two outputs from h^k

The item $F_k(\text{id}) = H_2(\text{"item"}, \text{id}, h^k) \bmod t$, for matching, and the label key $K(\text{id}) = H_2(\text{"label"}, \text{id}, h^k) \bmod t$, for opening labels. t = 65537.

Phase 1 by hand: O's polynomials

Phase 1 · O

O hashes its rows with k, pads each position with the row's label key, and interpolates mod t = 65537

τ(j)	id	item y_j	label key K	label $\ell_j = \tau + K$
1	150	35815	56830	**56831**
2	7e5 (dummy)	35081	13134	**13136**
3	221	39362	39074	**39077**
4	088	40782	18573	**18577**
5	399	52946	43556	**43561**
6	104	20938	60918	**60924**

Highlighted: the rows R's queries will match. O does not know this. $y_j = F_k(id)$, all mod t.

Reading a label

ℓ_j is the position padded with the row's label key. Only a party that ran the OPRF on that id has K(id) and can strip the pad: for 104, 60924 − 60918 = 6. The real protocol encrypts the label under a key derived from $F'_k(id)$; the additive pad is its toy stand-in.

Two polynomials mod t

$$P_b(X) = \prod_{j \in b} \left(X - F_k(\mathrm{id}_j)\right)$$

$$L_b\left(F_k(\mathrm{id}_j)\right) = \mathrm{Enc}_{F'_k(\mathrm{id}_j)}\left(\tau(j)\right)$$

P is zero exactly at O's six items. L passes through every (y_j, ℓ_j). Coefficients, lowest degree first:

X^m	P	L
1	24787	14745
X	46847	48338
X^2	41420	11025
X^3	31575	14990
X^4	62333	48363
X^5	37224	48931
X^6	1	—

One bin b here. Real runs hash items into thousands of bins, each with its own small P and L.

Phase 1 by hand: query and answer

Phase 1 · R ↔ O

Shaded values travel inside BFV ciphertexts under R's own key: O computes on them without seeing them

id	$q = F_k(id)$	q^2	q^4	P(q)	ρ (mask)	ρ·P(q)	L(q)	R's reading
104	20938	22851	34922	0	30623	**0**	60924	**match: τ = 60924 − 60918 = 6**
221	39362	6827	11122	0	5232	**0**	39077	**match: τ = 39077 − 39074 = 3**
317	12628	14863	49079	3913	24282	**52353**	39994	no match
9d1 (dummy)	58249	29974	59480	47288	42170	**40661**	34907	no match

Blue: sent by R. Grey: inside O's computation, never decrypted. Orange: returned to R.

1 R sends

Each item, encrypted under R's own BFV key, as q, q^2, q^4 (windowing). One item per query here; real runs cuckoo-hash the items into bins, at most one per bin.

2 O evaluates

$q^3 = q \cdot q^2$, $q^5 = q \cdot q^4$, $q^6 = q^2 \cdot q^4$ cost one multiplication. Then P(q) and L(q) from its coefficients, and P(q) times a fresh random $\rho \neq 0$. All on ciphertexts.

3 R decrypts

ρ·P(q) = 0 means a match: 104 and 221. Then τ = L(q) − K(id) mod t. For 317 and the dummy, ρ·P(q) is a random nonzero number and L(q) is meaningless.

Why ρ: without it, P(q) at a non-match would leak information about O's items.

Phase 1 · R

Phase 1: R computes its own block

A uses only R's rows in I, so R computes it exactly; it never leaves R

R's rows $\dot{x}_i = (1, x_{R1}, x_{R2})$ for $i \in I$

	intercept	education	health
$\dot{x}_{104}$	1.0	0.5	−0.2
$\dot{x}_{221}$	1.0	−0.4	0.8

$A = \dot{x}_{104}\,\dot{x}_{104}^{T} + \dot{x}_{221}\,\dot{x}_{221}^{T}$

	intercept	education	health
intercept	2.00	0.10	0.60
education	0.10	0.41	−0.42
health	0.60	−0.42	0.68

Check

A equals the blue block of G on slide 4: the protocol gets R's part exactly, with no encryption and no noise.

Why it matters here

A has rank 2 (two people, three columns), so its smallest eigenvalue is 0. That will decide where the ridge goes in stage 1.

Phase 2: upload

Phase 2 · R → O

R writes its data into O's shuffled order, zero everywhere it has no match, and encrypts it

	pos 1	pos 2	pos 3	pos 4	pos 5	pos 6
b (selection)	0.0	0.0	1.0	0.0	0.0	1.0
$\dot{x}_0$ (intercept)	0.0	0.0	1.0	0.0	0.0	1.0
$\dot{x}_1$ (education)	0.0	0.0	−0.4	0.0	0.0	0.5
$\dot{x}_2$ (health)	0.0	0.0	0.8	0.0	0.0	−0.2

Positions 3 and 6 are π(221) and π(104). $\dot{x}_0$ equals b: the intercept is 1 exactly where there is a match.

What O receives

- Enc(b): looks random
- Enc($\dot{x}_0$): looks random
- Enc($\dot{x}_1$): looks random
- Enc($\dot{x}_2$): looks random

R : $c_0 = -a\,s_R + e + \mathrm{Ecd}(v), \quad a = \mathrm{PRG}(\mathrm{seed})$ O : $c_0 \leftarrow c_0 - a\,s_O$

R encrypts each vector with its own share; O folds in its share to move it under the joint key. Every position looks the same to O, so it cannot see that 3 and 6 are special.

Phase 3 · O

Phase 3: aggregate

O multiplies R's encrypted vectors by its own columns, position by position, and sums

O's columns in its shuffled order

	1: 150	2: dummy	3: 221	4: 088	5: 399	6: 104
age x_{o1}	0.1	0.0	0.8	0.3	−0.2	−0.6
benefits x_{o2}	1.0	0.0	0.0	1.0	0.0	1.0
income y	−0.2	0.0	0.4	−0.5	0.9	0.7

Two of the sums, worked out (other positions multiply by 0)

$C_{11} = \Sigma_t \dot{x}_{1,t} \cdot age_t = (-0.4)(0.8) + (0.5)(-0.6) = \mathbf{-0.62}$ (cross-party: pairs R's row with O's row)

$B_{11} = \Sigma_t b_t \cdot age_t^2 = (0.8)^2 + (-0.6)^2 = \mathbf{1.00}$ (O's matched rows only)

All results, still encrypted

B: 1.00, −0.60, 1.00
C: 0.20, 1.00; −0.62, 0.50; 0.76, −0.20
c_R: 1.10, 0.19, 0.18
c_O: −0.10, 0.70
y^Ty: 0.65

Identical to the pooled answer.

Phase 3: packing the results

Phase 3 · O

Each released number sits alone in its own slot of a packed ciphertext

M_1: for stage 1

B and C only (9 slots); the other 8,183 slots hold 0

B_{11}	B_{12}	B_{22}	C_{01}	C_{02}	C_{11}	C_{12}	C_{21}	C_{22}	0 …
1.00	−0.60	1.00	0.20	1.00	−0.62	0.50	0.76	−0.20	

M_2: for stage 2

all 15 O-dependent numbers; y^Ty included, since it is free

B_{11}	B_{12}	B_{22}	C_{01}	C_{02}	C_{11}	C_{12}	C_{21}	C_{22}	c_{R0}	c_{R1}	c_{R2}	c_{O1}	c_{O2}	y^Ty
1.00	−0.60	1.00	0.20	1.00	−0.62	0.50	0.76	−0.20	1.10	0.19	0.18	−0.10	0.70	0.65

Two separate ciphertexts, so the first release can only ever depend on B and C. Slots run 1 to N/2 = 8,192 in the real scheme.

Phase 4: the release, on one number

Phase 4 · O → R

Toy arithmetic: one integer instead of a polynomial, modulus q = 1,000,003, no security

	Step	Formula	Value
	The statistic	$z = C_{11}$	−0.62
	Encode at scale Ω = 1000	$\mathrm{Ecd}(z) = \mathrm{round}(\Omega\, z)$	−620
	Aggregated ciphertext (key $s = s_R + s_O = 3 + 5$)	$c_0 = -a\, s + e + \mathrm{Ecd}(z)$, $c_1 = a$	(211,552, 723,481)
O	Re-randomize ($u = 1$, $e_0 = 1$, $e_1 = -2$)	$c_0' = c_0 + p_0\, u + e_0$, $c_1' = c_1 + a\, u + e_1$	(423,724, 446,957)
O	Add noise inside its share	$h_O = c_0' + c_1'\, s_O + \eta$, $\eta = -64{,}449$	−405,949
O	Send; keep c_0' secret	(c_1', h_O)	(446,957, −405,949)
R	Add its share	$v = h_O + c_1'\, s_R$	−65,081
R	…which equals	$\mathrm{Ecd}(z) + e_{tot} + \eta = -620 - 12 - 64{,}449$	−65,081
R	Decode	v / Ω	**−65.08**

R gets −0.62 plus noise of −64.46: the noise η/Ω has standard deviation $\sigma_2 = 37.5$. Without c_0', R cannot strip η off, and it never sees O's share.

Stage 1: choosing the ridge

Phases 4–5 · O → R

A first release of B and C only, with noise $\sigma_1 = 90.5$

	B_{11}	B_{12}	B_{22}	C_{01}	C_{02}	C_{11}	C_{12}	C_{21}	C_{22}
true	1.00	−0.60	1.00	0.20	1.00	−0.62	0.50	0.76	−0.20
released	**−0.5**	**55.9**	**73.1**	**18.6**	**−64.4**	**−72.8**	**120.2**	**118.5**	**−90.4**

R forms $\hat{G}_1$ = A ⊕ noisy B, C; its smallest eigenvalue	$\lambda_{min}(\hat{G}_1) = -219.6$
Radius for the stage-1 noise	$r_1 = \sigma_1(2\sqrt{5} + 2\sqrt{\ln 400}) = 847.3$
Target for stage 2	$g^* = 2 \cdot 2.5 \cdot \sigma_2 \cdot \sqrt{5} = 419.8$
Ridge on everything	$\lambda_I = \max\{0, g^* + r_1 - \lambda_{min}(\hat{G}_1)\} = 1486.8$
Ridge on O's block only	$\lambda_O = \infty$: it can never lift λ_{min} above $\lambda_{min}(A) = 0$
Choice	**Ψ = I, λ = 1487**

The ridge is huge because the noise is huge relative to two people's data. It is chosen before stage 2, so it is independent of the noise there.

Stage 2: the main release

Phase 4 · O → R

All fifteen O-dependent numbers, with noise $\sigma_2 = 37.5$

	B_{11}	B_{12}	B_{22}	C_{01}	C_{02}	C_{11}	C_{12}	C_{21}
true	1.00	−0.60	1.00	0.20	1.00	−0.62	0.50	0.76
released	**10.8**	**17.1**	**84.6**	**6.7**	**−4.5**	**−36.8**	**30.2**	**−66.5**

	C_{22}	c_{R0}	c_{R1}	c_{R2}	c_{O1}	c_{O2}	y^Ty
true	−0.20	1.10	0.19	0.18	−0.10	0.70	0.65
released	**−6.0**	**25.0**	**34.8**	**8.4**	**33.2**	**−2.3**	**77.6**

Gate

$\hat{\vartheta} = \lambda_{min}(\hat{G}_2 + \lambda I) / (2\sigma_2\sqrt{5}) = 8.42 \geq 1$, so R may report an estimate.

What R holds now

Its exact A, and fifteen numbers that are almost pure noise. Nothing about any one person in O's register can be read from them beyond what $\varepsilon = 1$ allows.

Phase 5: the estimate and its intervals

Phase 5 · R

$\tilde{\theta} = (\hat{G}_2 + \lambda I)^{-1}\,\hat{c}_2$, with a parametric bootstrap for the noise

Coefficient	Protocol estimate $\tilde{\theta}$	Wald interval $\tilde{\theta} \pm 2.00 \cdot sd$	Ridge target θ_λ (true)	Covers?
intercept	0.017	[−0.035, 0.068]	0.0007	**yes**
education	0.024	[−0.027, 0.075]	0.0001	**yes**
health	0.007	[−0.044, 0.057]	0.0001	**yes**
age	0.023	[−0.028, 0.074]	−0.0001	**yes**
benefits	−0.002	[−0.050, 0.046]	0.0005	**yes**

Reading it

- $\lambda \approx 1487$ dwarfs G (entries about 1), so $\theta_\lambda \approx 0$ and $\tilde{\theta} \approx \hat{c}_2 / \lambda$: noise, shrunk
- The intervals are honest about it: they are as wide as the estimates, and cover the target
- The least-squares β does not exist here (G has rank 2)

The protocol fails safe

With two people, any answer that told R something about them would break privacy. So the estimate is shrunk to almost nothing, and the intervals say so.

What each party saw

O

- 4 blinded OPRF queries and encrypted PSI queries
- Four upload ciphertexts that look random
- The padded sizes: 4 queries, 6 rows
- **Not: which rows matched, R's features, or that R has three people**
- **Not: any decryption share from R**

R

- Its own data, and its exact block A
- That 104 and 221 are in the register, at positions 6 and 3; that 317 is not
- Two noisy releases ($\sigma_1 = 90.5$, $\sigma_2 = 37.5$) and the estimate $\tilde{\theta}$
- **Not: O's rows, O's key share, or the c_0' that would remove the noise**

Nobody ever computed O's parts of G and c in the clear: the pooled answer on slide 4 exists only in this deck.

The same steps with 20,000 people

Same bounds, same budget, so the same noise: $\sigma_1 = 90.5$ and $\sigma_2 = 37.5$ per released number

Matched people drawn from a population like the toy's: R's features and age uniform on [−1, 1], benefits 0 or 1, income linear in all four plus noise, clipped to [−1, 1].

Coefficient	True β	Least squares, pooled (non-private)	Protocol $\tilde{\theta}$	Wald interval ± 2.00·sd
intercept	0.10	0.096	**0.095**	± 0.012
education	0.30	0.294	**0.296**	± 0.012
health	−0.20	−0.199	**−0.193**	± 0.013
age	0.40	0.395	**0.391**	± 0.013
benefits	0.20	0.199	**0.202**	± 0.019

Stage 1 now

$\lambda_{min}(\hat{G}_1) = 3{,}893 > g^* + r_1 = 1{,}267$, so $\lambda = 0$: no ridge. Gate $\hat{\vartheta} = 22.6$.

Why it works now

The noise per number is unchanged, but the statistics grew with n ($G_{11} \approx 6{,}700$). The estimate lands within 0.01 of pooled least squares.

What the example shows

1. Linkage gives R its matches and their positions in O's shuffled register; O learns only the padded sizes
2. R writes its data into O's order with zeros elsewhere, so O's encrypted sums pick out exactly the matched rows
3. O adds the noise inside its share of the decryption: R sees each statistic only with noise of the planned size
4. With two people the noise dominates; the ridge shrinks the estimate to almost nothing, and the intervals say so
5. With 20,000 people the same steps, with the same noise, recover least squares to within 0.01